

MEDCORE HMS

Business Requirements Document

Modular Hospital Management System (BRD)

One software; departments and features delivered as installable packages.

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Contents

1. Document Control

Field	Detail
Document title	MedCore HMS — Business Requirements Document
Version	1.0
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Status	Draft for stakeholder review
Related documents	Technical Architecture, Roadmap & Tooling Guide (companion)

2. Executive Summary

MedCore HMS is a single hospital management software whose departments and features are delivered as installable packages. A facility installs a small mandatory core (identity, patient record, audit, billing engine, module loader) and then enables only the packages it needs — OPD, Lab, Pharmacy, Billing, IPD, ABDM, Patient Portal, and more. The product targets small and mid-size Indian clinics and hospitals, is built entirely on free and open-source tools, and is compliant by design with India's DPDP Act and the ABDM digital-health standards. The package model doubles as the commercial model: customers pay for the core plus only the modules they turn on.

3. Business Objectives

- Digitise the full patient journey (registration → consultation → lab → pharmacy → billing, plus inpatient) for facilities that today run on paper.
- Sell a single, affordable product that fits a one-doctor clinic and a 50-bed hospital from the same codebase, via per-package enablement.
- Be ABDM-native and DPDP-compliant so AB-PMJAY-empanelled facilities can adopt it without compliance risk.
- Keep total cost of ownership low by using only free/open-source technology, passing the saving to customers as a competitive edge.
- Grow revenue as a hospital grows, by upselling additional packages without re-installation.

4. Project Scope

4.1 In scope

- Core platform + module framework (install/enable/disable packages).
- Clinical, pharmacy, billing, inpatient, ABDM/FHIR, and patient-facing packages (see Section 8).
- Staff web application, patient web portal and mobile app, and an installable desktop wrapper.
- Single-tenant on-premise deployment on each hospital's own server, with automated backups and central monitoring.
- DPDP, ABDM, AB-PMJAY-relevant, and GST compliance capabilities.

4.2 Out of scope (this phase)

- AI-generated medical advice or auto-diagnosis (explicitly excluded for safety/liability). Patient-facing content is limited to the treating doctor's own records and curated, doctor-attached material.
- Full PACS/DICOM imaging storage (a lightweight radiology package is planned; full PACS is future).
- Insurance-claims/TPA automation beyond ABDM record exchange (future package).
- Multi-tenant SaaS hosting (a future option once several installs exist).

5. Stakeholders & User Roles

Role	Description / key needs
Hospital Admin	Configures the system, manages users, enables packages, views analytics and revenue.
Doctor	Views patient history, records diagnosis, orders labs, writes prescriptions, discharges patients.
Nurse (IPD)	Records vitals and nursing notes; manages bed/ward tasks.
Lab Staff	Receives test orders, uploads results.
Pharmacy Staff	Manages stock/batches, dispenses against prescriptions.
Billing Staff	Aggregates charges, collects payment, issues GST receipts.
Patient	Views own records, prescriptions, lab results; books appointments; accesses bills (web + app).
System Vendor (you)	Installs, updates, monitors, and supports each deployment.

6. Business Process Flows

6.1 Outpatient (OPD) flow

1. Patient arrives; help desk registers or looks up the patient (UHID/ABHA).
2. Appointment/token issued; patient sees the doctor.
3. Doctor records diagnosis (ICD-10 coded), orders labs if needed, writes a prescription.
4. Lab runs tests and uploads results; doctor reviews.
5. Pharmacy dispenses against the prescription; stock auto-deducts.
6. Billing aggregates consultation + lab + pharmacy charges; patient pays; GST receipt issued.
7. Records (and FHIR bundle, if ABDM enabled) linked to the patient's ABHA.

6.2 Inpatient (IPD) flow

8. Doctor advises admission; a bed/ward is allocated (ADT: admit).
9. Nursing captures vitals and notes; orders, labs, pharmacy continue during the stay.
10. Room and service charges accrue to the patient's running bill.
11. On discharge (ADT: discharge), a discharge summary is generated and the final bill settled.

7. Non-Functional Requirements (NFRs)

ID	Category	Requirement
NFR-1	Security	RBAC enforced at API layer; encryption at rest and in transit; append-only audit log of all patient-data access.
NFR-2	Privacy (DPDP)	Consent capture; data export/erase; least-privilege access; data stays on the hospital's server.
NFR-3	Performance	Common screens respond in under 2 seconds for up to 100 concurrent staff users on a modest server.
NFR-4	Availability	Runs on the hospital LAN independent of internet; target 99.5% uptime; auto-restart on crash.
NFR-5	Backup/DR	Automated nightly encrypted backups with tested restore; pre-update backup + rollback.
NFR-6	Scalability	Handles a single-doctor clinic to a 50-bed hospital by enabling packages; vertical scaling on the server.
NFR-7	Usability	Role-based dashboards; regional-language UI; A4 + thermal printing; minimal training needed.
NFR-8	Maintainability	Modular packages, versioned independently; one-command install; central monitoring across installs.
NFR-9	Interoperability	ABDM/FHIR R4 record exchange; coded clinical data (ICD-10/LOINC/SNOMED).
NFR-10	Compliance	DPDP, ABDM, GST-correct billing; audit-ready.

8. Functional Requirements by Package

Requirements are grouped by package. The CORE is mandatory; all others are optional and enabled per hospital. Each requirement has an ID (FR-<pkg>-<n>).

8.1 Core Platform (mandatory)

ID	Requirement
FR-CORE-1	Authenticate users with role-based access (admin, doctor, nurse, lab, pharmacy, billing, patient).
FR-CORE-2	Maintain a single patient master record with UHID and optional ABHA number/address.
FR-CORE-3	Record an append-only audit log of every access or change to patient data.
FR-CORE-4	Capture and store patient consent with timestamp (DPDP).
FR-CORE-5	Provide a billing-engine ledger that all billing-related packages build upon.
FR-CORE-6	Install, enable, disable, and update packages via a Module Registry & Loader with dependency checks.
FR-CORE-7	Expose a single API gateway that mounts the routers of enabled packages only.

8.2 OPD / Consultation

Depends on: Core

ID	Requirement
FR-OPD-1	Register and search patients; issue/print OPD slips.
FR-OPD-2	Record diagnosis with ICD-10 codes, chief complaint, and notes.
FR-OPD-3	Create prescriptions with structured medicine, dose, frequency, and duration.
FR-OPD-4	Print prescriptions on A4 and thermal formats.

8.3 Appointments & Queue

Depends on: Core

ID	Requirement
FR-APPT-1	Book, reschedule, and cancel appointments per doctor.
FR-APPT-2	Issue and display token/queue numbers.

8.4 Lab & Imaging

Depends on: OPD

ID	Requirement
FR-LAB-1	Order lab tests (LOINC-coded) from a consultation.
FR-LAB-2	Upload and store results with reference ranges; notify the ordering doctor.
FR-LAB-3	Feed lab charges into billing.

8.5 Pharmacy & Inventory

Depends on: Billing Engine

ID	Requirement
FR-PHARM-1	Maintain a medicine master with HSN code and GST rate.
FR-PHARM-2	Track stock by batch and expiry; auto-deduct on dispense.
FR-PHARM-3	Raise low-stock and near-expiry alerts.
FR-PHARM-4	Feed dispensed medicine costs into billing.

8.6 Advanced Billing & Payments

Depends on: Billing Engine

ID	Requirement
FR-BILL-1	Auto-aggregate consultation, lab, pharmacy, and room charges into an itemised, GST-correct bill.
FR-BILL-2	Accept cash, UPI-QR, and optional card/UPI-gateway payments.
FR-BILL-3	Generate PDF receipts (A4 + thermal).
FR-BILL-4	Produce daily and monthly revenue and outstanding-payment reports.

8.7 IPD / Inpatient

Depends on: OPD, Billing

ID	Requirement
FR-IPD-1	Manage wards and beds with live occupancy status.
FR-IPD-2	Support admission, discharge, and transfer (ADT).
FR-IPD-3	Capture nursing notes and vitals.
FR-IPD-4	Accrue room/service charges to the patient bill; generate a discharge summary.

8.8 ABDM / FHIR Exchange

Depends on: Patient Master, OPD

ID	Requirement
FR-ABDM-1	Create/verify ABHA at registration; register facility (HFR) and providers (HPR).
FR-ABDM-2	Emit consultations, prescriptions, lab reports, and discharge summaries as valid FHIR R4 bundles.
FR-ABDM-3	Manage patient consent for health-information sharing (Consent Manager).

8.9 Patient Portal & Mobile

Depends on: OPD

ID	Requirement
FR-PORTAL-1	Let a patient securely log in (OTP/ABHA) on web and mobile.
FR-PORTAL-2	Show the patient ONLY their own diagnoses, prescriptions, lab results, and visit history.
FR-PORTAL-3	Present the treating doctor's care instructions in plain and regional language; allow download/print.
FR-PORTAL-4	Allow appointment booking and access to bills/receipts.
FR-PORTAL-5	Optionally show curated, doctor-attached educational material (never AI-generated medical advice).

8.10 Analytics & Dashboards

Depends on: any data package

ID	Requirement
FR-ANALYTICS-1	Show patient flow, revenue, doctor productivity, and inventory metrics to admins.

8.11 Localisation / Specialty

Depends on: Core / OPD

ID	Requirement
FR-LOCALE-1	Provide full regional-language UI and print.
FR-LOCALE-2	Provide specialty templates (e.g. dental, ophthalmology) layered on OPD.

9. Data Requirements (High-Level Entities)

Core-owned entities are shared by all packages; each package owns its own entities in its own database schema.

Core: patients, users, doctors, departments, facility, audit_log, consent_artifacts, bill (ledger).

OPD: appointments, consultations, prescriptions, prescription_items.

Lab: lab_orders, lab_results.

Pharmacy: medicines, stock_batches, pharmacy_dispensing.

Billing: bills, bill_items.

IPD: wards, beds, admissions, nursing_notes.

ABDM: fhir_exports, consent references (ABHA/HFR/HPR ids on core records).

10. Integration Requirements

- ABDM (NHA) — ABHA, HFR/HPR, Consent Manager, FHIR R4 exchange (free sandbox, then production certification).
- Payments — optional UPI/card gateway SDK; UPI-QR requires no integration cost.
- SMS/notifications — optional provider for OTP and alerts (a paid gateway may apply; not a software cost).

11. Assumptions, Constraints & Dependencies

- Assumption: each hospital provides a server meeting the minimum spec (see companion technical guide).
- Constraint: only free/open-source tools; unavoidable third-party fees (payment/SMS) are borne per transaction.
- Constraint: single-tenant on-premise for launch; data resides on the hospital's server.
- Dependency: ABDM production go-live depends on NHA certification timelines.
- Dependency: at least one design-partner facility must be secured before full build (validation).

12. Acceptance Criteria (High-Level)

- Core: a package can be installed, enabled, and disabled without affecting other packages or data.
- OPD MVP: a full outpatient visit can be completed end-to-end and printed.
- Compliance: every data access is audit-logged; ABDM sandbox flows validate against the India FHIR profile.
- Ops: a fresh server can be provisioned to a running, backed-up install in under one hour.

13. Key Risks

Risk	Mitigation
Building before validating demand	Secure a design-partner and pilot the OPD MVP early.
ABDM certification delay	Start sandbox work early; ABDM is an isolated package so delays don't block others.
Support load across many installs	Identical installs, central monitoring, automated backups; per-package versioning.
Data-loss / breach	Encryption, tested backups, rollback, DPDP controls from the core.

14. Glossary

Term	Meaning
ABDM	Ayushman Bharat Digital Mission — India's national digital-health framework.
ABHA	Ayushman Bharat Health Account — a citizen's health ID under ABDM.

Term	Meaning
AB-PMJAY	Pradhan Mantri Jan Arogya Yojana — national health-insurance scheme.
ADT	Admission, Discharge, Transfer — the inpatient movement lifecycle.
DPDP	Digital Personal Data Protection Act, 2023 (India).
FHIR	Fast Healthcare Interoperability Resources — HL7 data-exchange standard.
HFR / HPR	Health Facility Registry / Healthcare Professional Registry (ABDM).
RBAC	Role-Based Access Control.
UHID	Unique Health Identification number (facility-level).

This BRD is a planning artifact, not legal advice. DPDP/ABDM/AB-PMJAY/GST requirements evolve — verify current obligations with the NHA and a compliance advisor before production.